



Discrete Probability Distributions · Worksheet 1

Further Statistics 1 Chapter 1: Discrete Probability Distributions. Answer all questions. Show your working clearly. Total: 89 marks. Time: 110 minutes.

Name _____

Class _____

Date _____

INSTRUCTIONS

- Answer **all** questions in the spaces provided. Show all your working.
- There are four sections, one for each topic in Chapter 1. Within each section the questions increase in difficulty; the **last question** of each section is set at A level exam standard.
- Give exact answers (fractions) where appropriate; otherwise give probabilities to 4 decimal places and other answers to 3 significant figures.
- The **Roar-level challenge** at the end is a harder stretch question.
- The marks for each question are shown in brackets, e.g. [3].

Section	Marks available	Marks scored
A Expected value	20	
B Variance	21	
C Functions of a random variable	21	
D Constructing probability distributions from a context	22	
Roar-level challenge	5	
Total	89	

Section A Expected value

20 marks

1. The discrete random variable X has the probability distribution shown.

[2]

x	1	2	3	4
$P(X = x)$	0.1	0.3	0.4	0.2

Find $E(X)$.

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2. The discrete random variable X has the probability distribution shown, where k is a constant. [3]

x	0	1	2	3
$P(X = x)$	0.2	k	$2k$	0.2

Find the value of k and find $E(X)$.

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3. The discrete random variable X has probability function [3]

$$P(X = x) = \frac{x}{15}, \quad x = 1, 2, 3, 4, 5.$$

Show that the probabilities sum to 1, and find the exact value of $E(X)$.

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4. The discrete random variable X has the probability distribution shown, where a and b are constants. [4]

x	1	2	3	4
$P(X = x)$	a	b	0.3	0.2

Given that $E(X) = 2.3$, find the value of a and the value of b .

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5. The discrete random variable X has probability function

$$P(X = x) = \begin{cases} k(5 - x) & x = 1, 2, 3, 4 \\ k & x = 5 \\ 0 & \text{otherwise} \end{cases}$$

where k is a constant.

(a) Show that $k = \frac{1}{11}$. [2]

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(b) Find the exact value of $E(X)$. [2]

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(c) Find $P(X > E(X))$. [2]

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(d) Two independent observations X_1 and X_2 are taken of X . Find $P(X_1 + X_2 = 3)$. [2]

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(Total for Question 5 is 8 marks)



Section B Variance

21 marks

6. The discrete random variable X has the probability distribution shown.

[2]

x	0	1	2
$P(X = x)$	0.25	0.5	0.25

Given that $E(X) = 1$, find $\text{Var}(X)$.

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7. For the random variable X in Question 1, find $E(X^2)$, $\text{Var}(X)$ and the standard deviation of X .

[3]

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8. The random variable X is the score when a fair six-sided dice is rolled.

[3]

Find $E(X)$ and show that $\text{Var}(X) = \frac{35}{12}$.

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9. The discrete random variable X has probability function

[4]

$$P(X = x) = \frac{x}{10}, \quad x = 1, 2, 3, 4.$$

Find $E(X)$ and $\text{Var}(X)$.

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10. The discrete random variable X has the probability distribution shown, where p and q are constants.

x	1	2	3	4
$P(X = x)$	0.2	p	q	0.3

Given that $E(X) = 2.7$,

(a) find the value of p and the value of q , [4]

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(b) find $\text{Var}(X)$, [3]

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(c) find $P(|X - \mu| < \sigma)$, where μ and σ are the mean and standard deviation of X . [2]

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(Total for Question 10 is 9 marks)



Section C Functions of a random variable

21 marks

11. The random variable X has $E(X) = 4$ and $\text{Var}(X) = 3$. [2]
Find $E(2X + 5)$ and $\text{Var}(2X + 5)$.

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12. The random variable X has $E(X) = 2$ and $\text{Var}(X) = 1.5$. [3]
Find (a) $E(3 - 2X)$, (b) $\text{Var}(3 - 2X)$, (c) $E(X^2)$.

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13. The discrete random variable X has the probability distribution shown. [3]

x	1	2	3
$P(X = x)$	0.2	0.5	0.3

The random variable $Y = X^2 - 2X$. Find $E(Y)$.

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14. The random variable X has $E(X) = 3$ and $\text{Var}(X) = 2$. The random variable $Y = aX + b$, where a [4]
and b are constants and $a > 0$.
Given that $E(Y) = 10$ and $\text{Var}(Y) = 18$, find the value of a and the value of b .

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15. The discrete random variable X has the probability distribution shown.

x	1	2	3	4
$P(X = x)$	0.4	0.3	0.2	0.1

(a) Find $E(X)$. [2]

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(b) Show that $\text{Var}(X) = 1$. [3]

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(c) Find $E(5 - 2X)$ and $\text{Var}(5 - 2X)$. [2]

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(d) Find $E(2^X)$. [2]

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(Total for Question 15 is 9 marks)



Section D Constructing probability distributions from a context

22 marks

16. A fair coin is tossed three times. The random variable X is the number of heads. [2]

Find the probability distribution of X .

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17. Two fair six-sided dice are rolled. The random variable X is the larger of the two scores (if the scores are equal, X is that score). [3]

Find the probability distribution of X .

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18. A bag contains 3 red counters and 2 blue counters. Two counters are taken at random, without replacement. The random variable X is the number of red counters taken. [3]

Find the probability distribution of X and $E(X)$.

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19. A player pays £2 to play a game in which a fair six-sided dice is rolled once. The player receives £6 for a six, £3 for a four or a five, and nothing otherwise. [4]

The random variable X is the player's profit, in pounds. Find the probability distribution of X and show that the game is fair.

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20. In a game, a biased coin with $P(\text{head}) = 0.4$ is tossed until a head appears, or until it has been tossed three times, whichever happens first. The random variable X is the number of times the coin is tossed.

(a) Find the probability distribution of X . [3]

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(b) Find $E(X)$. [2]

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(c) Find $\text{Var}(X)$. [3]

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(d) A player receives $\pounds(9 - 3X)$. Find the entry fee that makes the game fair. [2]

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(Total for Question 20 is 10 marks)



ROAR-LEVEL CHALLENGE

5 marks

21. The discrete random variable X takes only the values 0, 1 and 2. Given that $E(X) = 1$ and $\text{Var}(X) = 0.6$, find the probability distribution of X . [5]

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END OF WORKSHEET

Total: 89 marks